

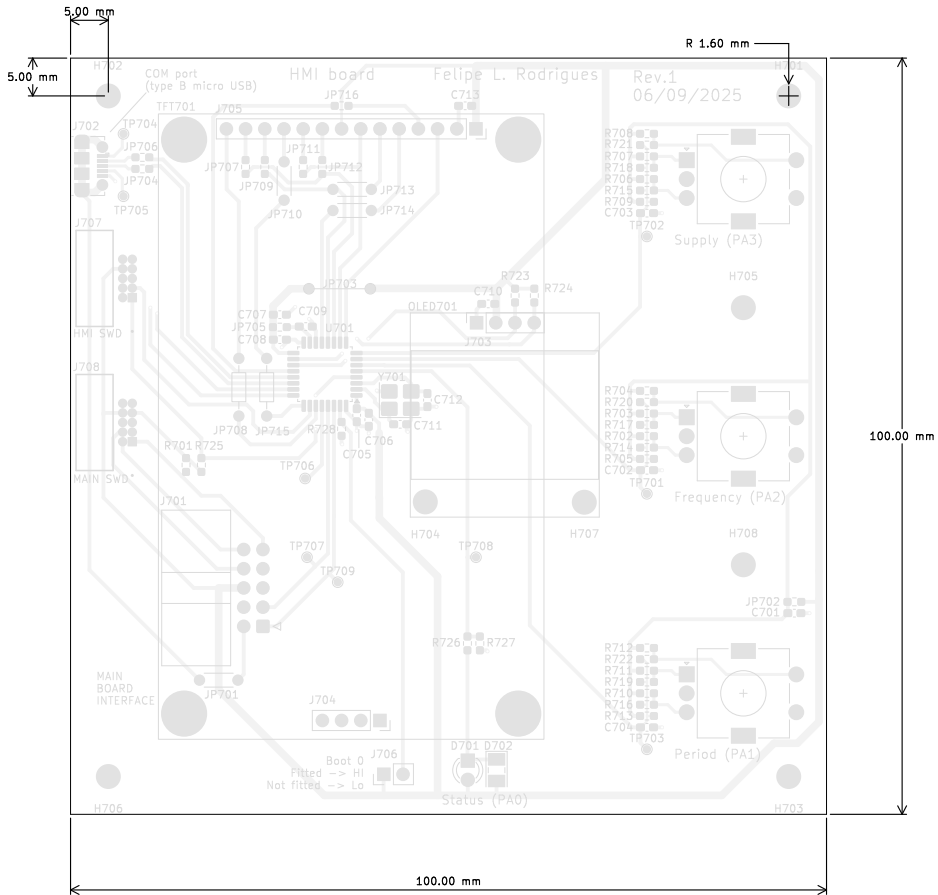
HMI board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen			Direct Printing	Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	Core		1.48mm	FR4_7628	Dielectric	
	Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL LEAD-FREE
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE 100.000 x 100.000 mm
BOARD THICKNESS 1.660 mm
TRACE WIDTH 0.300 mm
TRACE TO TRACE 0.180 mm
MIN. HOLE (PTH) 0.300 mm
MIN. HOLE (NPTH) 3.000 mm
ANNULAR RING 0.100 mm
COPPER TO HOLE 0.250 mm
COPPER TO EDGE 0.500 mm
HOLE TO HOLE 0.250 mm



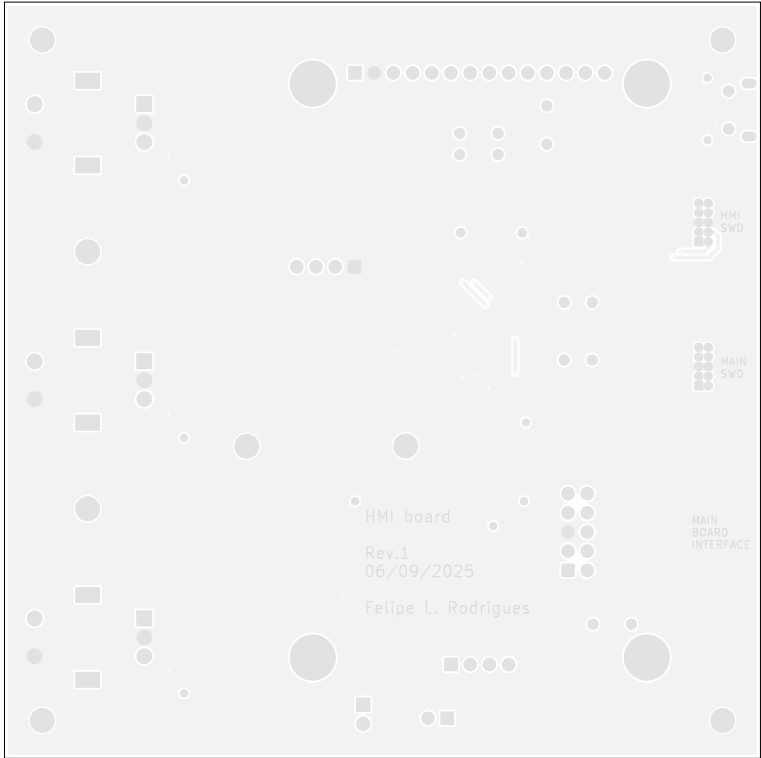
Comments:	Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
	Board Name: HMI board		Project Name: 1 kV strobe	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13
Sheet Path:		Reviewer:	Size: A3	Revision: + (Unreleased)
			Sheet: 1 of 12	

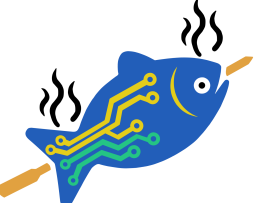
	1	2	3	4	5	6	7	8
	HMI board Fabrication Document							Bottom Fabrication (Scale 1:1)
A								
B								
C								
D								
E								
F								

HMI board Fabrication Document

Bottom Fabrication (Scale 1:1)

(BOTTOM SIDE VIEW)



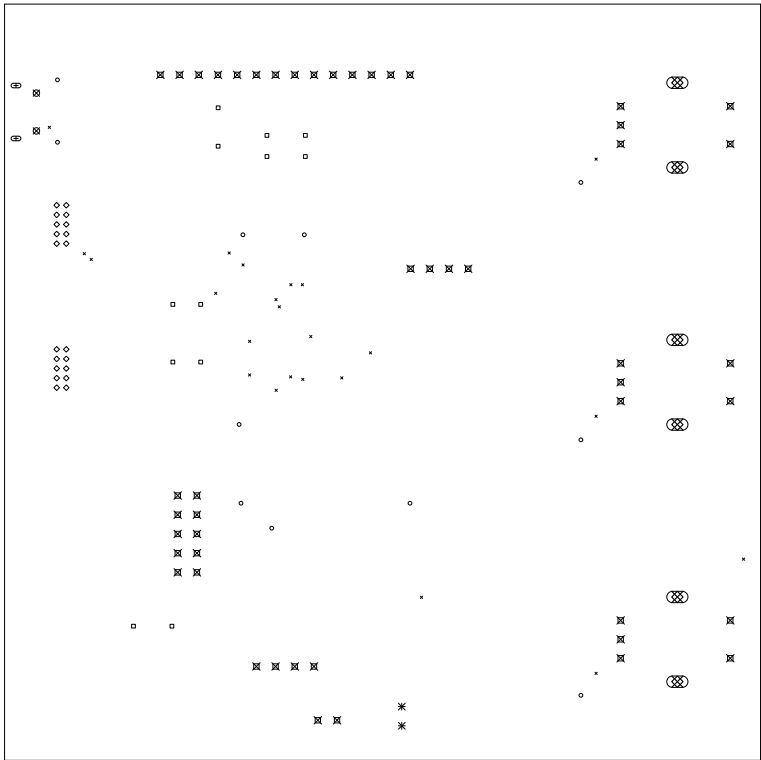
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	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:	Reviewer:		Size: A3	Sheet: 2 of 12

HMI board Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	23	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Via
○	11	0,50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
+	2	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
□	12	0,70mm (27,56mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
◇	20	0,74mm (29,13mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⊗	2	0,85mm (33,46mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
※	2	0,90mm (35,43mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⊗	49	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
⊗	6	1,50mm (59,06mils)	PTH	Slot	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
Total 127						



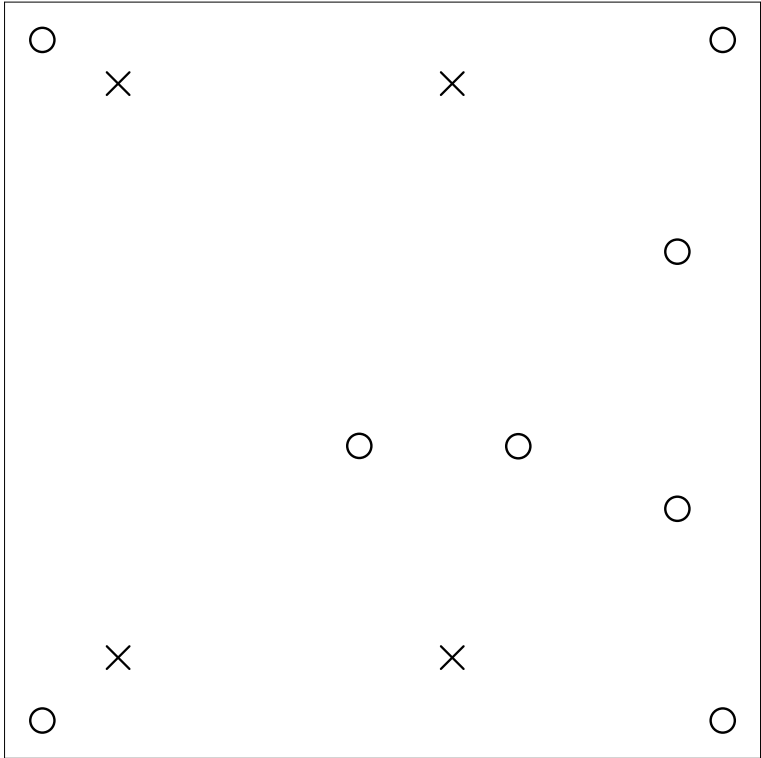
	Comments:	Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
		Board Name: HMI board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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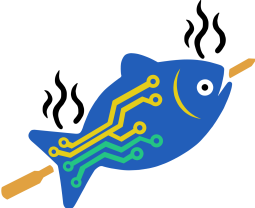
HMI board Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

Drill Table

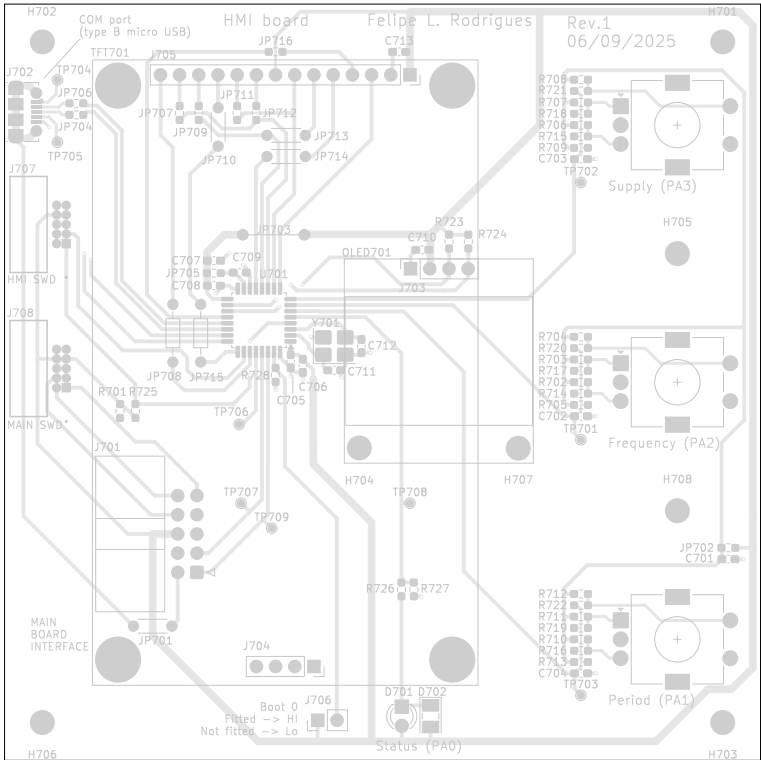
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	3.00mm (118,11mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Mechanical
○	8	3.20mm (125,98mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Mechanical
Total 12						



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	Sheet Title: Drill Drawing (L1 - L2)	File Name: Strobe_HMI.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:			Reviewer:	Size: A3	Sheet: 4 of 12

HMI board Fabrication Document

Top Test Points (Scale 1:1)



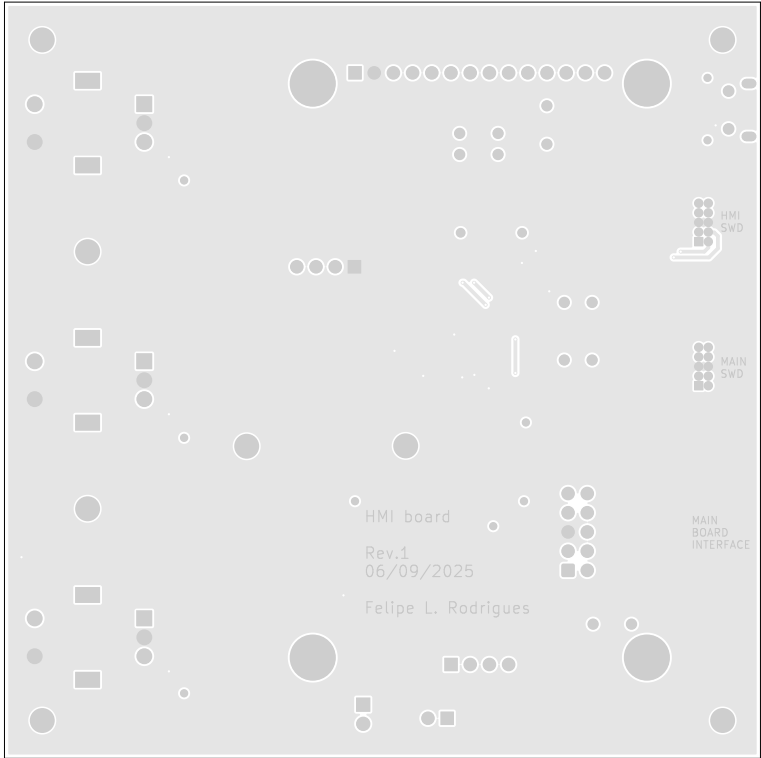
Ref.	Net	X [mm]	Y [mm]
TP701	FREQ	86.24	52.38
TP702	SUP	86.24	86.42
TP703	PER	86.24	18.60
TP704	Net-(J702-VBUS)	17.00	99.98
TP705	Net-(J702-ID)	17.00	91.73
TP706	Net-(U701-PB5)	41.03	54.41
TP707	LV_TX	41.28	44.00
TP708	ST_LED	63.63	44.00
TP709	LV_RX	45.34	40.70

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HMI board Fabrication Document

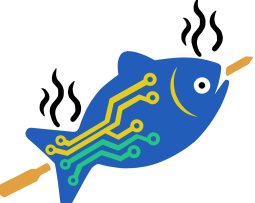
Bottom Test Points (Scale 1:1)

(BOTTOM SIDE VIEW)



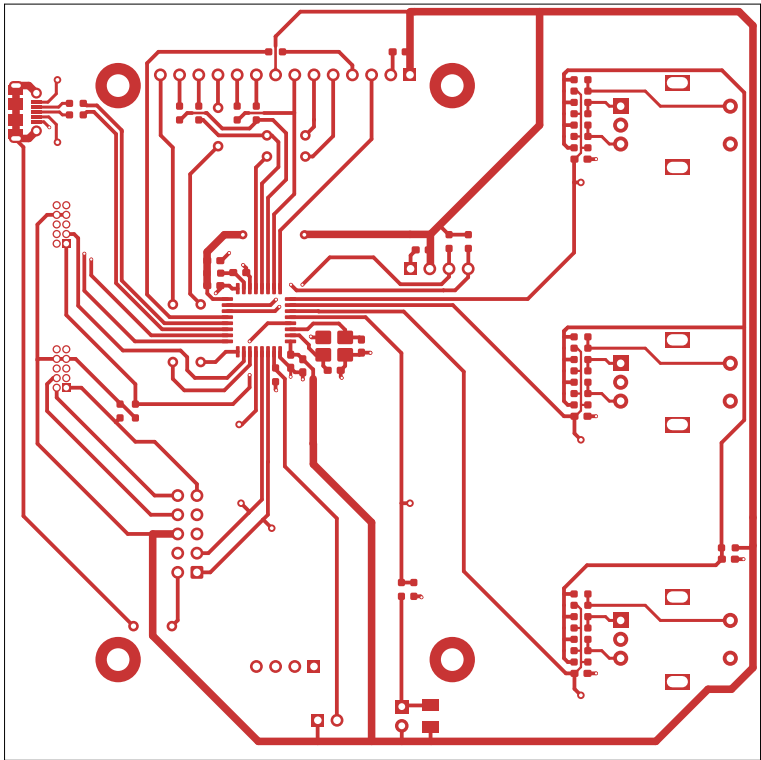
Ref.	Net	X [mm]	Y [mm]
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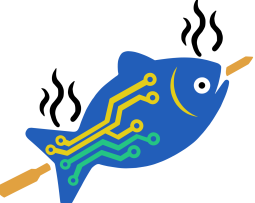


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	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
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HMI board Fabrication Document

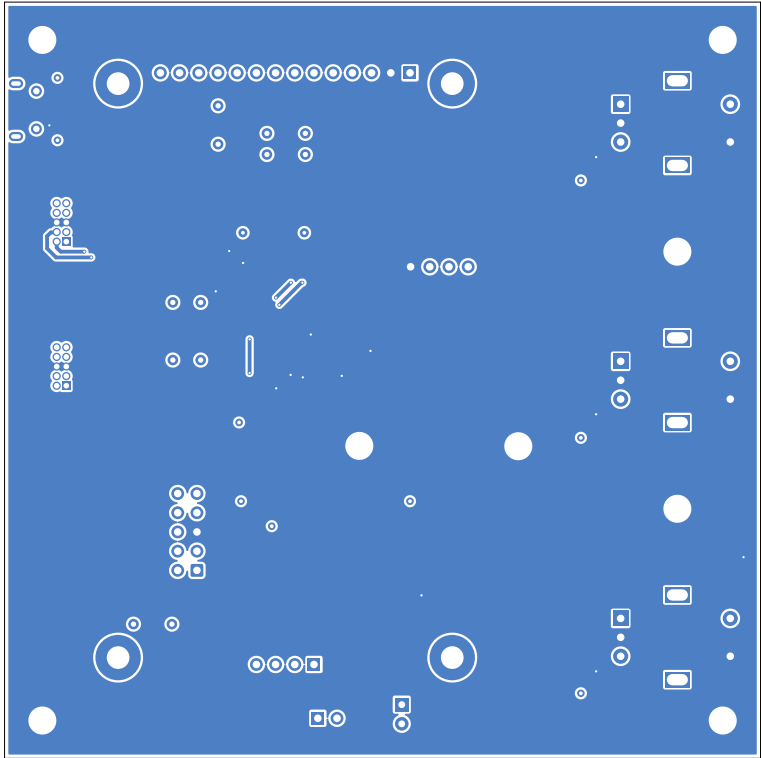
L1 (Sig, PWR) (Scale 1:1)

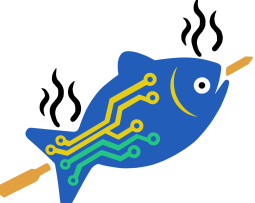


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	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 7 of 12
Project Name: 1 kV strobe					

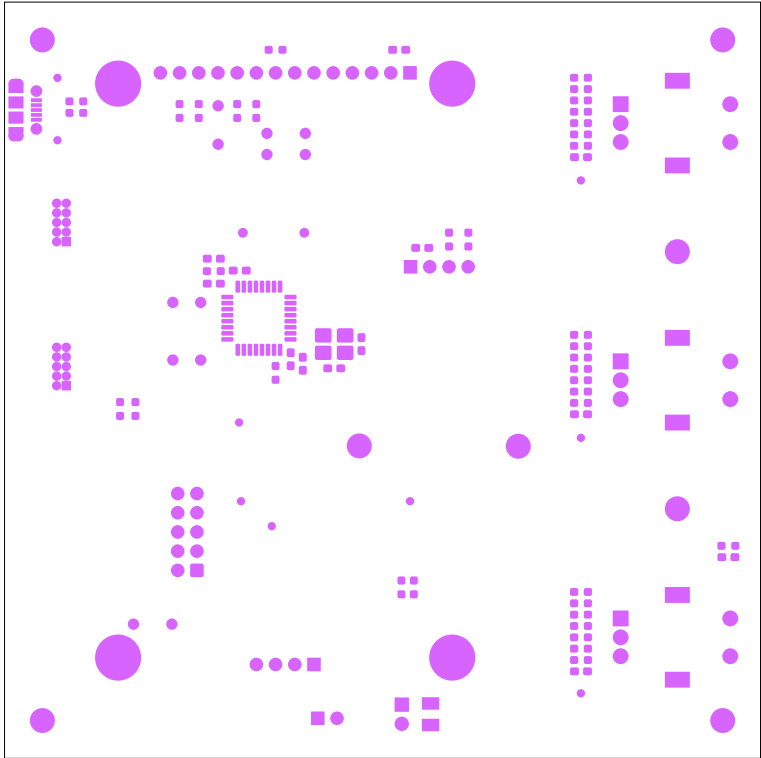
HMI board Fabrication Document

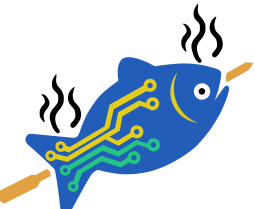
L2 (Sig, PWR) (Scale 1:1)

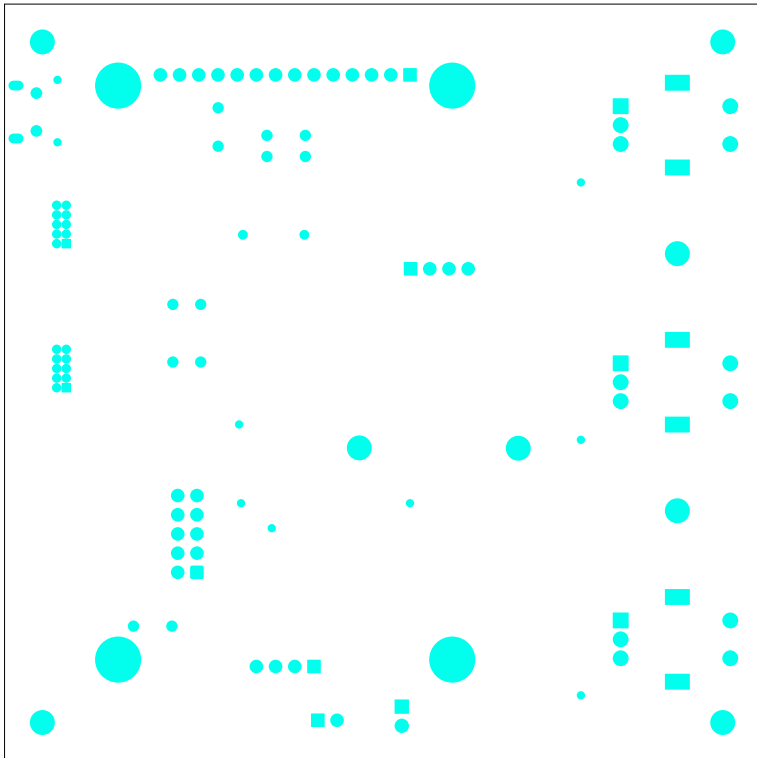
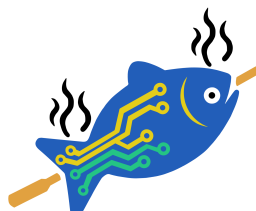


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	Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 12

	1	2	3	4	5	6	7	8
	HMI board Fabrication Document							F.Mask (Scale 1:1)
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8



 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
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	Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 12

HMI board Fabrication Document				B.Mask (Scale 1:1)			
							
 ChurrasCod Electronics		Comments:		Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
		Sheet Title: B.Mask (Scale 1:1)		Board Name: HMI board		Project Name: 1 kV strobe	
				File Name: Strobe_HMI.kicad_pcb		Designer: FR	Date: 2024-04-13
Sheet Path:				Reviewer:		Size: A3	Sheet: 10 of 12

HMI board Fabrication Document

B.Silkscreen (Scale 1:1)

Felipe L. Rodrigues
Rev. 1
08\09\2022
HMI Board Interface
MAIN BOARD
2W2
2W2

 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
			Board Name: HMI board		Project Name: 1 kV strobe	
	Sheet Title: B.Silkscreen (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 12 of 12	

HMI board Fabrication Document

B.Silkscreen (Scale 1:1)

Felipe L. Rodrigues
Rev. 1
08\09\2022

HMI board
MAIN BOARD
INTERFACE

2W2
DIN

2W2
HMI

 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: CHECKED	Git Hash: d96861b
			Board Name: HMI board		Project Name: 1 kV strobe	
	Sheet Title: B.Silkscreen (Scale 1:1)	File Name: Strobe_HMI.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: + (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 12 of 12	

 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: CHECKED		Git Hash: d96861b	
			Board Name: HMI board		Project Name: 1 kV strobe			
	Sheet Title: B.Silkscreen (Scale 1:1)		File Name: Strobe_HMI.kicad_pcb		Designer: FR		Date: 2024-04-13	Revision: + (Unreleased)
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